

The exam is open books, notes, printed materials. You are allowed web access. Consulting/discussing with anyone which includes asking questions in any forum during the exam is strictly prohibited. *Unless specified explicitly, assume 64-bit X86 architecture with multiple CPUs.*
All the best!

1. Select the correct answer(s) for the following questions. Note that, you have to *write all correct choices* to get the credits (no partial marks). No explanation required.

 $3 \times 5 = 15$

- (a) Which of the following C/assembly statement(s) *may* result in a user-to-kernel mode switch? Assume that all addresses are virtual addresses and R1, R2 are general purpose registers.

A. `malloc(1024);`
 B. `MOV R1, R2; // Move the value in register R1 to register R2`
 C. `MOV R1, (R2); // Move the value in register R1 to the memory address in R2`
 D. `POP R1; // Move the value at top-of-stack address to R1`
 E. `RET; // Return from the current function`

- (b) Consider that there is a page fault for virtual address V of a multi-threaded process. When the page fault handler inspects the page table mappings, it finds that the page table is correct for all the levels except for the leaf level (e.g., L4 PTE in a 4-level page table). In the leaf-level page table entry,

A. the present bit can be zero
 B. the present bit can be one
 C. the write bit can be zero (read-only access allowed)
 D. the write bit can be one (both read-write allowed)
 E. everything can be correct in all aspects

- (c) Assuming TLB hardware without support for address space identifiers (ASID), which of the following case(s) a TLB flush (fully or selectively) is a must? Assume split-mode address space layout where the OS consumes a part of the process address space.

A. When any user process is scheduled after handling an interrupt
 B. When a user process invokes a system call
 C. When a physical page frame corresponding to a virtual address is swapped out
 D. When a physically mapped virtual address is unmapped using `unmap` system call
 E. Context switch between two threads of a same process

- (d) Consider the following code.

```
sem_t S1, S2, S3; /* All initialized to zero */
string(){
    wait(S1);
    printf("A");
    wait(S2);
    printf("B");
    wait(S3);
    printf("C");
}

number(){
    printf("1");
    post(S1);
    printf("2");
    post(S2);
    printf("3");
    post(S3);
}
```

Which of the following can be a correct output if one thread executes `string()` and the other thread executes `number()` concurrently?

A. 1A2B3C
 B. 12ABC3
 C. 123ABC
 D. 12ABC3
 E. 12A3BC

- (e) Which of the following file related system calls can be served without performing any disk I/O before the system call returns to user space? Assume a file system with block-layer caching and successful completion of all the system calls.

A. `open`
 B. `open` with `O_CREAT`
 C. `lseek`
 D. `read`

E. write

2. True or False with justification. Justification should be a precise argument supporting your verdict. No marks without proper reasoning.

$2 \times 5 = 10$

- (a) A process is created only when CPU is executing in the kernel-mode.
- (b) Two threads of the same process can correctly communicate with each other without invoking any system calls.
- (c) LRU page replacement always results in less number of page faults than FIFO page replacement.
- (d) If a file system always performs synchronous writes, there will be no consistency issues.
- (e) In non-preemptive scheduling, a process can be replaced by another process when it is executing an infinite **for** loop in user mode.